Grade 9 Natural Sciences

Exam study notes · Internal paper (1 hour)
Scientific investigation · Human reproduction · The Earth as a system · The atmosphere · Stars

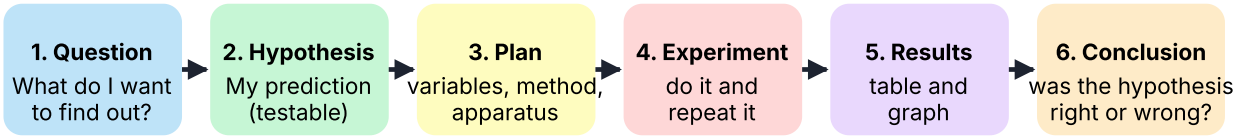
Contents

1. Scientific investigation: hypothesis, variables, reliability, validity, graphs
2. Human reproduction (textbook pp. 35–42)
3. The Earth as a system: spheres, interactions, the lithosphere, the rock cycle (pp. 197–210)
4. Mining: mineral resources, ore, refining, mining in South Africa (pp. 197–210)
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6. The birth, life and death of stars (pp. 217–221)
7. Definitions to learn by heart

Your guideline also says to work through the textbook activities on these pages.

1. Scientific investigation

Scientists answer questions by doing **investigations** – planned experiments that test an idea in a fair way. Every investigation follows the same steps.



The steps of a scientific investigation

Hypothesis

A **hypothesis** is a **testable prediction** about how one variable affects another. It says what you **think** will happen, before you test it. Write it like this:

> **"If** [I change the independent variable], **then** [the dependent variable will ...]."
> *Example:* "If a plant gets **more hours of sunlight**, then it will **grow taller**."

A hypothesis can turn out to be **wrong** – that is fine. The experiment tells you whether to **accept** or **reject** it.

Variables

A **variable** is anything that can change in an experiment. There are three kinds.

Variable	What it is	Easy way to remember	Plant example
Independent variable	The one thing **you change on purpose**	"I change it"	Hours of sunlight
Dependent variable	What you **measure** – it changes **because** of the independent variable	It "depends" on what I changed	Height of the plant
Controlled variables	Everything you **keep the same**, so the test is fair	"Keep the same"	Type of plant, amount of water, soil, pot size, temperature

> Change only **one** variable (the independent variable) at a time. If you changed two things, you couldn't tell which one caused the result.

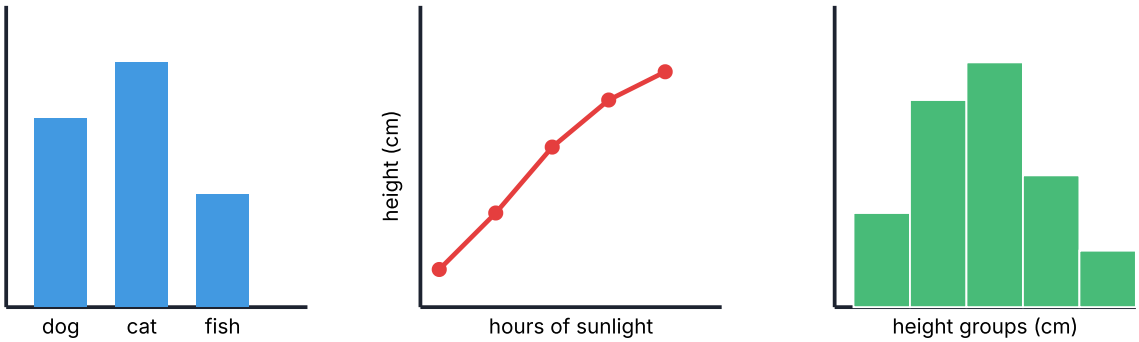
Reliability and validity

	Meaning	How to make an investigation ...
Reliability	You would get the **same results** if you (or someone else) did the experiment again	**Repeat** the experiment several times (or use several plants) and work out the **average**
Validity	The experiment really **tests what it is supposed to test** – it is a **fair test**	Change only the independent variable, keep all controlled variables the same, and measure accurately with the right instruments

> **Reliable** = **repeatable** (same results every time). **Valid** = **fair** (testing the right thing, only one variable changed).

Graphs

Results are recorded in a **table** and then shown on a **graph**. The type of graph depends on the type of data.



The three types of graphs

Graph	Use it when...	Example
Bar graph	The independent variable is in **separate groups (categories)**. Bars have **gaps** between them.	Favourite pets in the class; rainfall in different towns
Line graph	Both variables are **numbers that change continuously**; shows a **trend** (pattern)	Plant height over hours of sunlight; temperature over time
Histogram	Numerical data is **grouped into ranges (intervals)** and you show how many are in each range (**frequency**). Bars **touch** – no gaps.	Number of learners in height groups 140–149 cm, 150–159 cm ...

Rules for drawing any graph:

1. Give it a **heading**: "A graph showing the relationship between [dependent variable] and [independent variable]".
2. The **independent variable** goes on the **x-axis** (horizontal); the **dependent variable** on the **y-axis** (vertical).
3. **Label both axes** with the **units**, e.g. "Height of plant (cm)".
4. Use an **even scale** (equal spaces for equal amounts) that uses most of the paper.
5. Plot points accurately; for a line graph join them with a line or draw a line of best fit.

Worked example of a full investigation

Part	Example
Investigative question	How does the number of hours of sunlight affect the growth of bean plants?
Hypothesis	If bean plants get more hours of sunlight, then they will grow taller.
Independent variable	Hours of sunlight per day (0, 2, 4, 6, 8 hours)
Dependent variable	Height of the plant after 2 weeks (cm)
Controlled variables	Type of plant, amount of water, type of soil, pot size, temperature
Reliability	Use 3 plants for each amount of sunlight and calculate the average height
Conclusion	The more hours of sunlight, the taller the plants grew. The hypothesis is **accepted**.

Hours of sunlight per day	**0**	**2**	**4**	**6**	**8**
Average height (cm)	2	5	9	12	14

This data is drawn as the **line graph** in both variables are numbers, so a line graph shows the trend.

2. Human reproduction

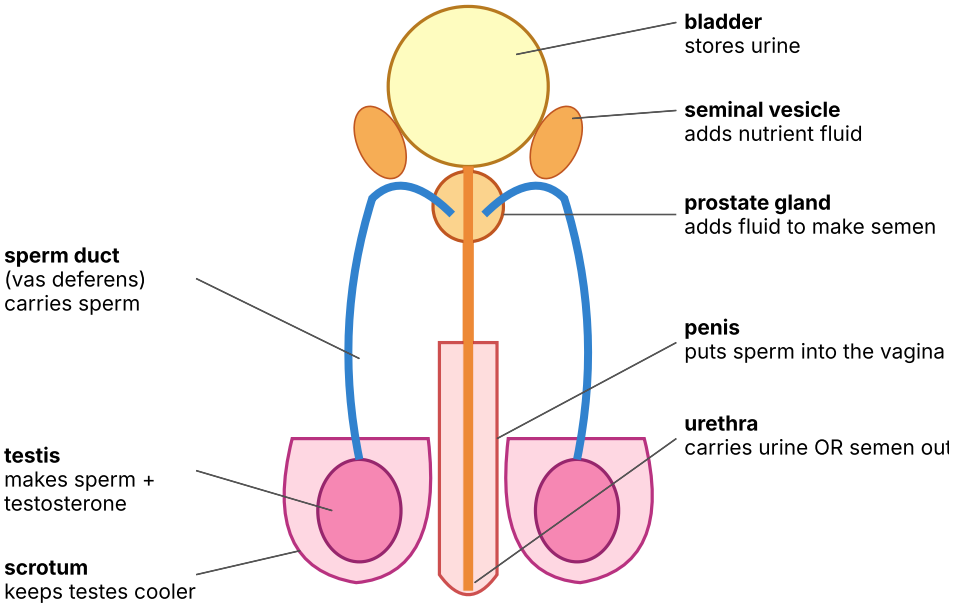
Reproduction is how living things produce offspring. Humans reproduce **sexually**: a male **sex cell (sperm)** joins a female **sex cell (egg / ovum)**. Sex cells are made in the **gonads** – the **testes** in males and the **ovaries** in females.

Puberty

Puberty is the stage (usually between 9 and 16 years) when the body becomes able to reproduce. It is controlled by **hormones**: **testosterone** in boys and **oestrogen** and **progesterone** in girls.

Changes in boys	Changes in girls	Both
Voice deepens Facial and chest hair Shoulders broaden, muscles develop Testes start making sperm Penis and testes grow	Breasts develop Hips widen Ovaries start releasing eggs Menstruation (periods) begins	Growth spurt Pubic and underarm hair Sweat more, body odour Pimples (acne) Emotional changes, mood swings

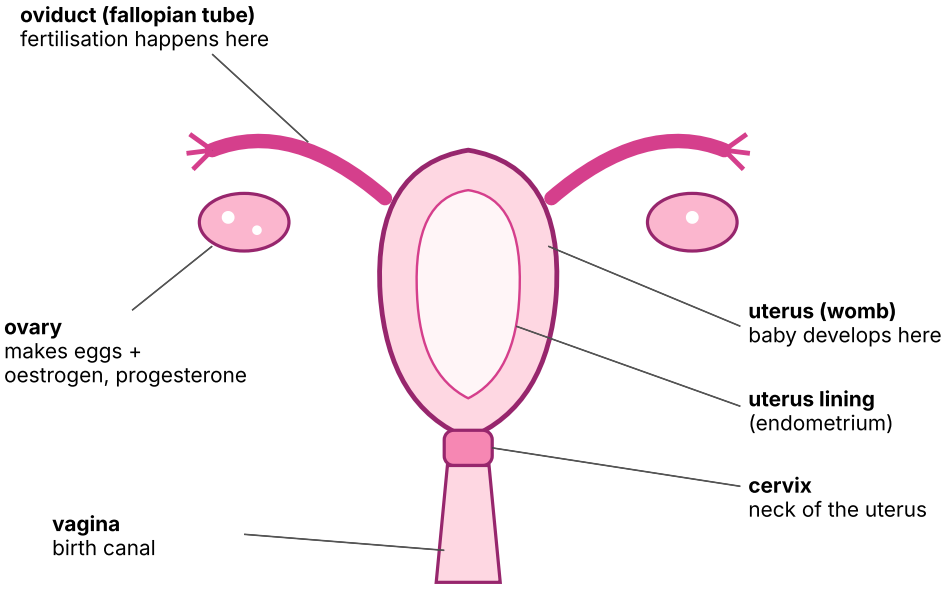
The male reproductive system



The male reproductive system (front view, simplified)

Path of sperm: testis → sperm duct → (seminal vesicle and prostate add fluid → **semen**) → urethra → out through the penis. Testes hang outside the body in the scrotum because sperm develop best slightly **below body temperature**.

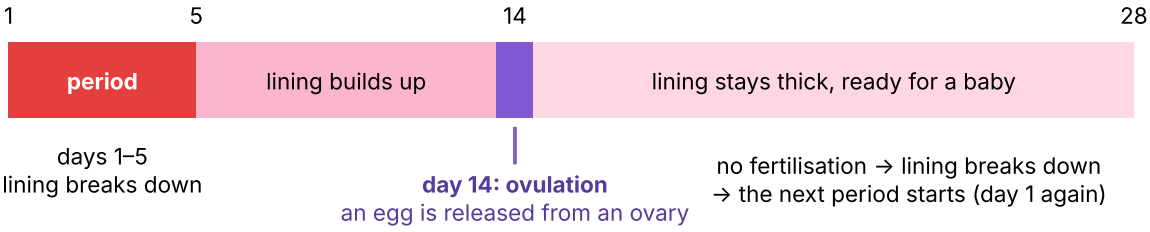
The female reproductive system



The female reproductive system (front view, simplified)

The menstrual cycle

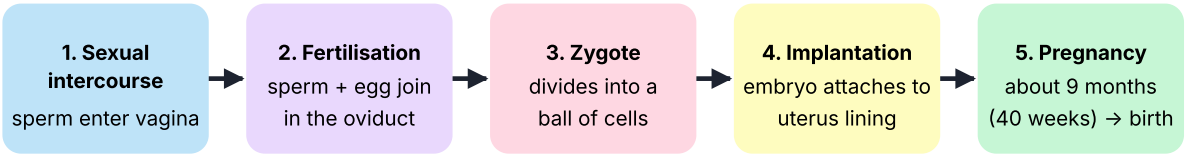
About every **28 days** the female body prepares for a possible pregnancy. This is the **menstrual cycle**.



The 28-day menstrual cycle

1. **Days 1–5 – menstruation (period):** the thick lining of the uterus breaks down and passes out through the vagina with some blood.
2. **Days 6–13:** the uterus lining **thickens** again (controlled by oestrogen). An egg ripens in an ovary.
3. **About day 14 – ovulation:** a mature egg is released from an ovary into the oviduct. This is when a woman is most likely to fall pregnant.
4. **Days 15–28:** the lining stays thick (progesterone). If the egg is **not fertilised**, it dies, the lining breaks down and a new cycle starts. If it **is** fertilised, there is no period – the woman is pregnant.

Fertilisation, pregnancy and birth



embryo (first 8 weeks) → foetus (from about 9 weeks until birth)

From fertilisation to birth

- **Fertilisation:** one sperm joins the egg in the **oviduct**. The fertilised egg is a **zygote**. It contains half its genetic material from the father and half from the mother.
- The zygote divides many times as it moves to the uterus and **implants** in the thick lining.
- The **placenta** forms: it lets food and oxygen pass from the mother's blood to the baby, and waste from the baby to the mother. The baby is joined to the placenta by the **umbilical cord**. The **amniotic fluid** protects the baby from bumps.
- Alcohol, smoking and drugs pass through the placenta and can harm the baby (e.g. foetal alcohol syndrome).
- **Birth:** the uterus muscles contract, the cervix opens and the baby is pushed out through the vagina.

Preventing pregnancy and STIs

Method	How it works
Abstinence	Not having sex – the only 100% effective method against pregnancy and STIs
Condom (male / female)	Barrier that stops sperm reaching the egg; also protects against STIs and HIV
Contraceptive pill / injection / implant	Hormones stop ovulation; do **not** protect against STIs
IUD (loop)	Small device in the uterus that stops implantation

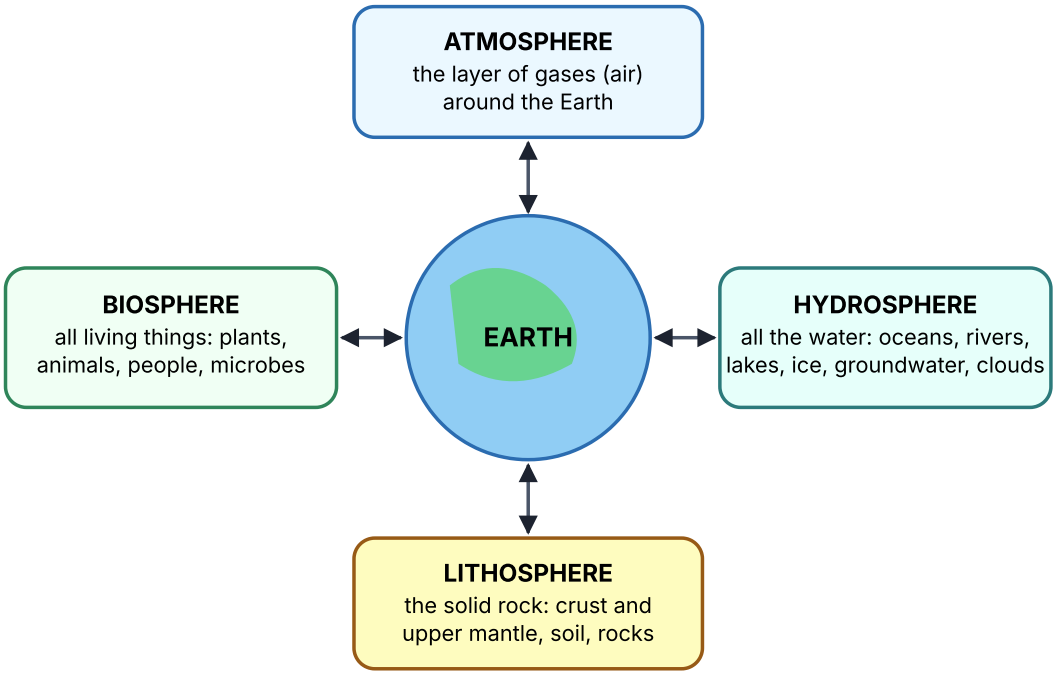
Sexually transmitted infections (STIs) spread through sexual contact: HIV, syphilis, gonorrhoea, chlamydia, herpes. Many have no symptoms at first. Prevent them with abstinence, condoms and testing; treat them early at a clinic.

> **Teenage pregnancy** can lead to dropping out of school, health risks for mother and baby, and money problems. Delay sex, get correct information, and talk to a trusted adult or clinic.

3. The Earth as a system

A **system** is a set of parts that work together and affect each other. The Earth is a system made of **four spheres**. A change in one sphere causes changes in the others.

The four spheres of the Earth



The four spheres all interact with each other

Sphere	What it includes	Greek root
Atmosphere	Layer of gases around the Earth: nitrogen 78%, oxygen 21%, argon about 0.9%, carbon dioxide about 0.04%, water vapour	*atmos* = vapour
Hydrosphere	All water on Earth – oceans (about 97%), ice caps and glaciers, rivers, lakes, groundwater, water vapour	*hydro* = water
Lithosphere	The rigid outer layer of the Earth: the **crust** and the top part of the **mantle** – rocks, minerals, soil, mountains	*lithos* = stone
Biosphere	All living organisms, and the parts of the other spheres where life exists	*bios* = life

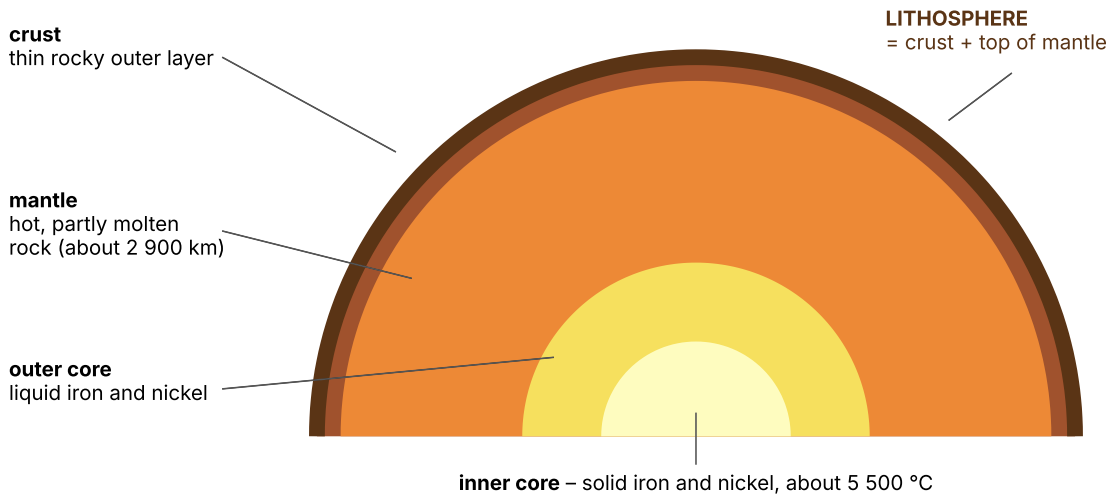
Interactions between the spheres

The spheres are always affecting each other. Exam questions often give an example and ask which spheres interact.

Example	Spheres that interact
Plants take in carbon dioxide and give off oxygen (photosynthesis)	Biosphere ↔ atmosphere
Animals breathe in oxygen and breathe out carbon dioxide	Biosphere ↔ atmosphere
Rain falls and erodes rocks and soil	Atmosphere / hydrosphere → lithosphere
A volcano erupts and releases gases and ash into the air	Lithosphere → atmosphere
Plant roots break rocks and help form soil	Biosphere → lithosphere
Animals drink water; fish live in the sea	Biosphere ↔ hydrosphere
Water evaporates from the sea and forms clouds	Hydrosphere → atmosphere
People mine minerals and burn coal, polluting the air	Biosphere → lithosphere and atmosphere

The water cycle is a good example: water evaporates from the sea (hydrosphere) into the air (atmosphere), falls as rain onto the land (lithosphere), and is used by plants and animals (biosphere).

The structure of the Earth and the lithosphere

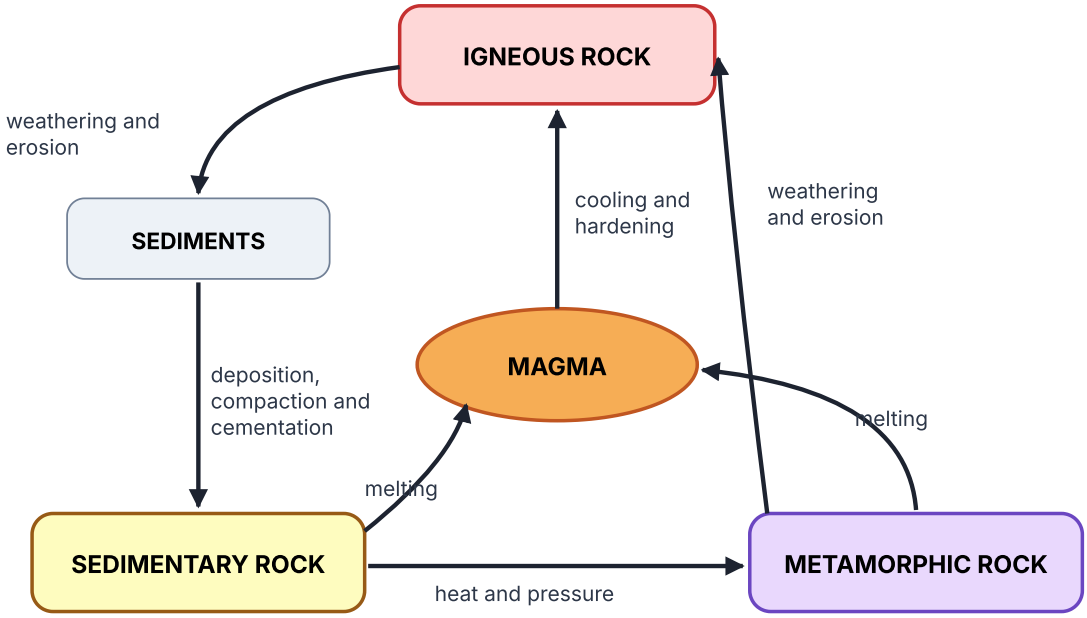


A cut-away of the Earth showing its layers

- The **crust** is the thin outer layer we live on: about **5–10 km** thick under the oceans and **30–70 km** under the continents.
- The **lithosphere** is the **crust plus the rigid upper part of the mantle**. It is broken into large pieces called **tectonic plates** that move very slowly.
- The lithosphere gives us **soil** for farming, **rocks and minerals** for building and mining, and **fossil fuels** (coal, oil, gas).

Rocks and the rock cycle

Rock type	How it forms	Examples
Igneous	**Magma** (molten rock) or lava **cools and hardens**. Fast cooling (on the surface) → small crystals; slow cooling (underground) → large crystals	Granite, basalt, pumice
Sedimentary	Rock is weathered and eroded into small pieces (**sediments**) that are **deposited in layers**, then **compacted** (squashed) and **cemented** together over millions of years. Can contain **fossils**.	Sandstone, shale, limestone, coal
Metamorphic	Existing rock is changed by great **heat and pressure** deep underground, without melting	Marble (from limestone), slate (from shale), quartzite (from sandstone)



Rocks keep changing from one type to another over millions of years – there is no start or end.

The rock cycle

The rock cycle is the continuous process in which rocks are formed, broken down and changed from one type into another by **weathering, erosion, deposition, compaction, heat, pressure, melting and cooling**.

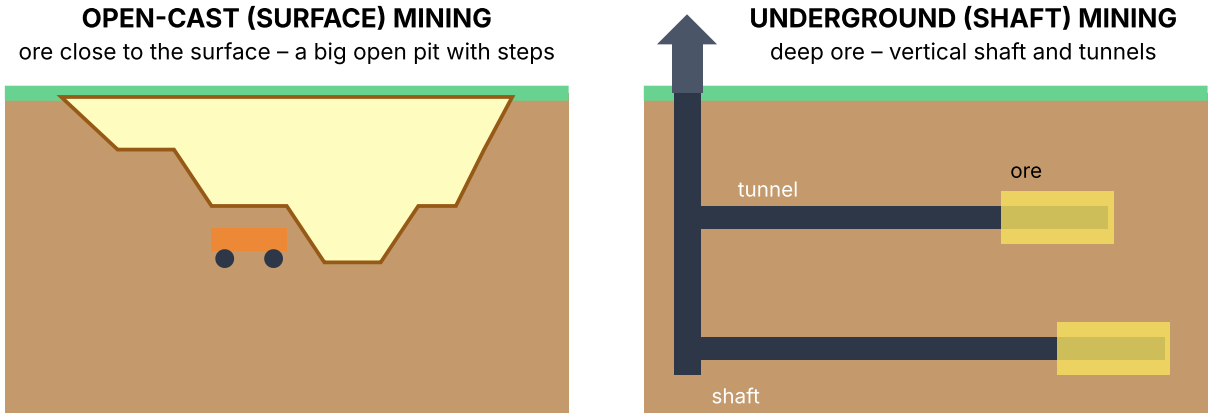
4. Mineral resources and mining

The lithosphere contains valuable **minerals** that we dig up (mine), process and use. South Africa is one of the richest countries in the world in minerals.

Key words

Word	Meaning
Mineral	A naturally occurring, non-living solid with a definite chemical make-up, found in rocks (e.g. gold, quartz, iron oxide)
Ore	A rock that contains **enough** of a valuable mineral to make it worth (profitable) to mine
Mineral resource	Minerals that people can use, e.g. metals, coal, diamonds
Extraction	Removing the ore or mineral from the ground (mining)
Refining	Purifying a mineral or metal by removing the unwanted parts
Non-renewable	Minerals took millions of years to form and will run out – they cannot be replaced

Extraction of mineral resources: mining methods



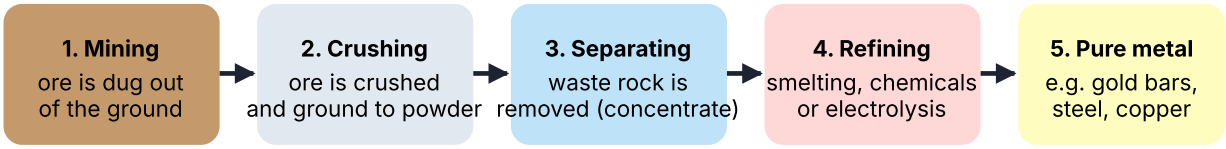
e.g. coal, iron ore, copper, diamonds

e.g. gold, platinum, deep coal

Open-cast mining and underground mining

Method	How it works	Used for
Open-cast (open-pit) mining	Soil and rock are removed to make a huge open pit; the ore near the surface is dug out with big machines and explosives	Coal, iron ore, copper, diamonds (e.g. the Big Hole in Kimberley)
Strip mining	Layers of soil are stripped away in long strips to reach ore lying in a flat layer just below the surface	Coal
Underground (shaft) mining	A vertical **shaft** is dug down, then horizontal **tunnels**; miners and machines bring the ore up in lifts	Gold, platinum (SA gold mines are among the deepest in the world – about 4 km)
Alluvial mining	Minerals are separated from sand and gravel in riverbeds or on the seabed	Diamonds along the west coast

From ore to pure metal: extraction and refining



The steps from ore to a pure metal

- **Crushing and grinding:** the ore is broken into tiny pieces so the mineral can be separated from the waste rock.
- **Separating (concentrating):** the valuable mineral is separated from the waste, e.g. by **froth flotation** (mineral sticks to bubbles), using its density (heavy particles sink), or using magnets (iron).
- **Refining** – making the metal pure:
  - **Smelting:** the ore is heated to a very high temperature in a furnace so that the metal melts and separates. *Iron* is smelted in a **blast furnace** with coke (carbon) and limestone.
  - **Chemical methods:** *gold* is dissolved out of crushed ore with a **cyanide** solution, then recovered and melted into bars.
  - **Electrolysis:** an electric current is used to separate and purify metals such as *copper* and *aluminium*.
- The waste rock is dumped on **mine dumps** (tailings).

Mining in South Africa

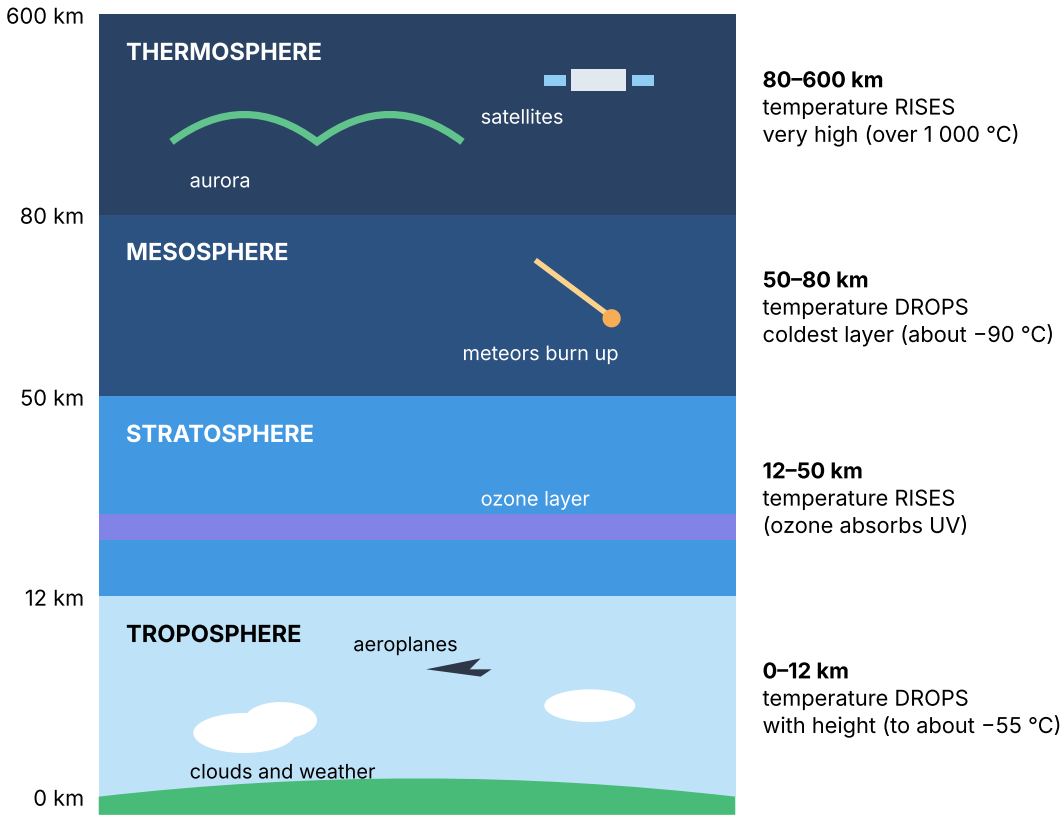
Mineral	Where in SA	Used for
Gold	Witwatersrand (Gauteng), Free State	Jewellery, electronics, money reserves
Platinum	Bushveld Complex (North West, Limpopo) – SA has most of the world's platinum	Catalytic converters in cars, jewellery, electronics
Coal	Mpumalanga, Limpopo	Electricity (power stations), steel making, fuel
Diamonds	Kimberley (Northern Cape), Cullinan (Gauteng), west coast	Jewellery, cutting and drilling tools
Iron ore	Sishen (Northern Cape), Limpopo	Making steel
Manganese, chrome	Northern Cape (Kalahari), Bushveld	Making stainless steel

Advantages (benefits) of mining	Disadvantages (problems) of mining
Creates **jobs** for many people Earns money from **exports** – important for the economy Provides raw materials for industry and electricity Roads, towns and services are built near mines	Destroys the land and natural habitats (holes, dumps) **Pollutes water** – e.g. **acid mine drainage** from old mines Dust and air pollution from mine dumps and smelters Dangerous for miners: rock falls, lung diseases (e.g. TB, silicosis) Minerals are **non-renewable** – they will run out

Mining companies must **rehabilitate** the land after mining: fill in holes, replace topsoil, plant vegetation and treat polluted water.

5. The atmosphere

The **atmosphere** is the layer of gases that surrounds the Earth, held in place by gravity. It gives us the oxygen we breathe, keeps the Earth warm, protects us from harmful UV rays and burns up meteors. It is divided into **four layers** based on how the **temperature changes** with height.



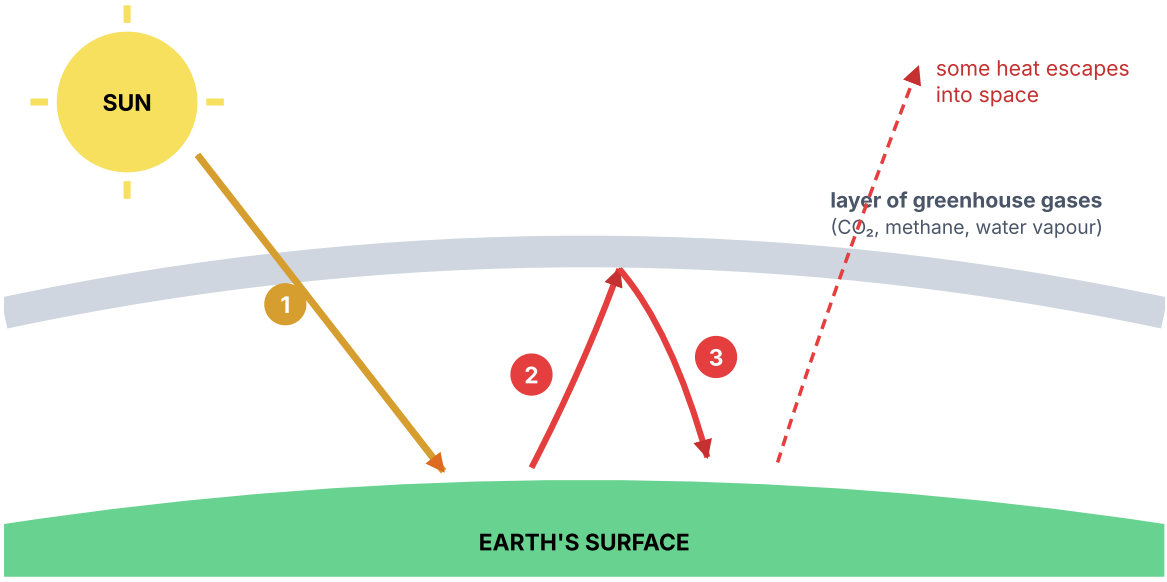
The four layers of the atmosphere (not to scale)

Layer	Height	Temperature	Important facts
Troposphere	0 to about 12 km (8 km at the poles, 16 km at the equator)	**Decreases** as you go up (about 6.5 °C less every km)	Closest to Earth; contains about **75–80% of the air** and almost all water vapour. **All weather** (clouds, rain, wind) happens here. We live here; planes fly at the top.
Stratosphere	12 to 50 km	**Increases** with height	Contains the **ozone layer**, which **absorbs harmful ultraviolet (UV) radiation** from the Sun – this is why it gets warmer. Very still, dry air; no weather.
Mesosphere	50 to 80 km	**Decreases** – the **coldest** layer (about −90 °C)	**Meteors (shooting stars) burn up** here because of friction with the air.
Thermosphere	80 to about 600 km	**Increases** – very high (over 1 000 °C)	Air is very thin. Absorbs X-rays and UV from the Sun. The **aurora** (northern and southern lights) happens here. The International Space Station and many **satellites** orbit here.

Memory aid (from the ground up): **T**he **S**trong **M**an **T**ires – Troposphere, Stratosphere, Mesosphere, Thermosphere. Temperature pattern going up: **down, up, down, up**.

The ozone layer is damaged by **CFCs** (chemicals once used in fridges and aerosol sprays), causing an **ozone hole**. More UV reaching Earth causes sunburn, skin cancer and eye damage. CFCs have now been banned.

The greenhouse effect



The greenhouse effect keeps the Earth warm. The numbers match the steps below.

1. Energy from the **Sun** passes through the atmosphere and **heats the Earth's surface**.

2. The warm Earth gives off **heat** (infrared radiation) back towards space.
3. **Greenhouse gases** – mainly **carbon dioxide (CO_2)**, **methane (CH_4)** and **water vapour** – absorb some of this heat and send it back to Earth. This **traps heat**, like the glass of a greenhouse.

> **The natural greenhouse effect is good:** without it, the Earth's average temperature would be about −18 °C instead of about 15 °C, and life could not survive.

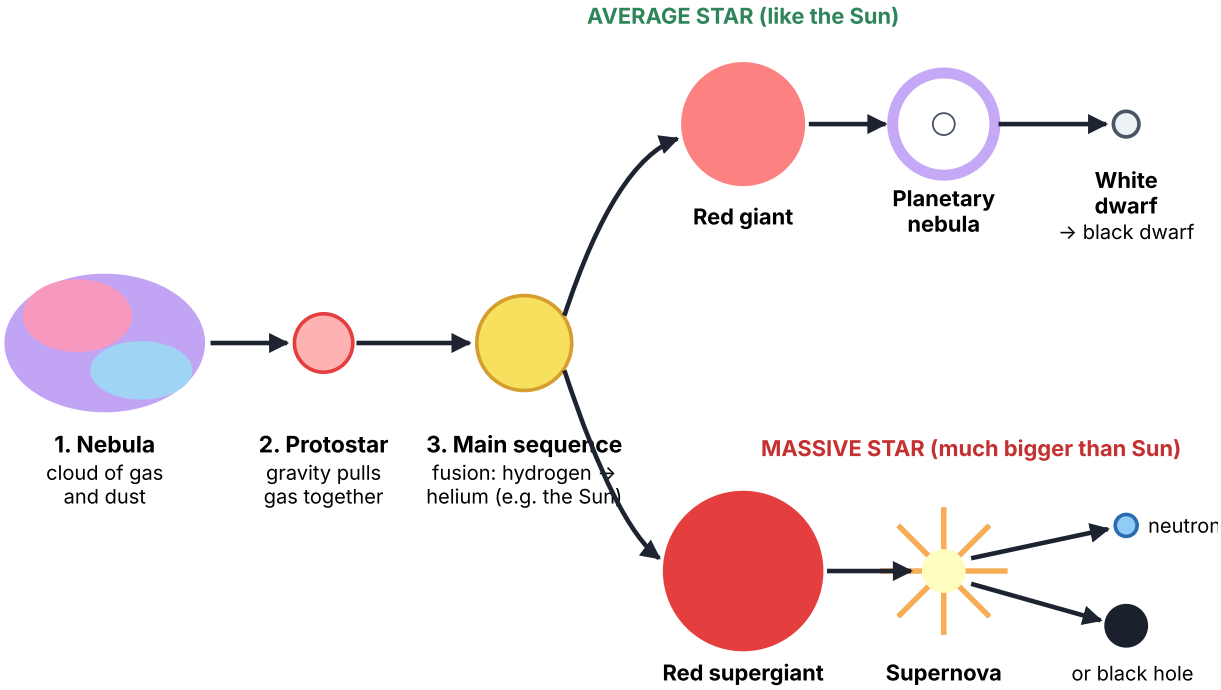
The enhanced greenhouse effect and global warming

Humans are adding **extra** greenhouse gases to the atmosphere, so **more heat is trapped** and the Earth is getting warmer. This is the **enhanced greenhouse effect**, which causes **global warming** and **climate change**.

Human causes	Effects of global warming	What we can do
Burning **fossil fuels** (coal, oil, petrol) in power stations, cars and factories → CO_2 **Deforestation** – fewer trees to take in CO_2 Farming cattle and rubbish dumps → methane	Ice caps and glaciers melt → **sea levels rise** More extreme weather: **droughts, floods, heat waves**, storms Plants and animals lose habitats; some become extinct Crop failures → food shortages	Use less electricity; use **renewable energy** (solar, wind) Use public transport, walk or cycle **Plant trees**; stop deforestation Reduce, reuse, recycle

6. The birth, life and death of stars

A **star** is a huge ball of very hot, glowing gas (mostly **hydrogen** and **helium**) that gives off its own light and heat. Our **Sun** is a medium-sized star. Stars are born, live for millions or billions of years, and then die. What happens to a star depends on its **mass** (how big it is).



The life cycle of stars. What happens after the main-sequence stage depends on the star's mass.

Birth of a star

1. A star begins in a **nebula** – a huge cloud of **gas (mostly hydrogen) and dust** in space.
2. **Gravity** pulls the gas and dust together into a spinning clump. As it gets squashed it heats up and forms a **protostar** (a baby star).
3. When the centre (core) gets hot enough – about 15 million °C – **nuclear fusion** starts: **hydrogen atoms join (fuse) to form helium**, releasing huge amounts of **light and heat**. A star is born.

Life of a star: the main sequence

- A **main-sequence star** is a stable star that is fusing hydrogen into helium in its core. Stars spend about **90% of their lives** in this stage.
- The star is stable because two forces are **balanced**: **gravity** pulls inwards and the **pressure of the hot gas** from fusion pushes outwards.
- The **Sun** is a main-sequence star. It is about **4.6 billion years old** and will stay like this for about another **5 billion years**.
- **Bigger stars burn faster**: they are hotter and brighter, but live much shorter lives. A star's colour shows its temperature: **blue = hottest**, then white, yellow (like the Sun), orange, **red = coolest**.

Death of a star

When the hydrogen in the core runs out, the star begins to die. How it dies depends on its mass.

Average (Sun-sized) star	Massive star (more than about 8 times the Sun's mass)
1. The star swells into a **red giant** – huge, cooler and red. (The Sun will grow big enough to swallow Mercury and Venus.) 2. Its outer layers drift off into space as a glowing shell of gas, a **planetary nebula**. 3. The hot core that is left is a **white dwarf** – small (about the size of the Earth) and very dense. 4. Over billions of years it cools and fades into a **black dwarf**.	1. The star swells into an enormous **red supergiant**. 2. The core collapses and the star explodes in a **supernova** – one of the biggest explosions in the universe, briefly brighter than a whole galaxy. 3. What is left becomes either a **neutron star** (tiny and extremely dense) or, for the biggest stars, a **black hole** – where gravity is so strong that **not even light can escape**.

Stars make the elements: fusion in stars and supernova explosions make heavier elements such as carbon, oxygen and iron. A supernova scatters these into space, where they form new nebulae, new stars and planets. The atoms in your body were made inside stars!

7. Definitions to learn by heart

These are the usual Grade 9 definitions for the topics in your guideline. If your teacher or textbook words a definition differently, learn their wording.

Term	Definition
Investigation	A planned experiment that tests an idea in a fair way.
Hypothesis	A testable prediction about how the independent variable will affect the dependent variable.
Independent variable	The variable that you change on purpose in an investigation.
Dependent variable	The variable that you measure; it changes because of the independent variable.
Controlled variables	The variables that are kept the same so that the test is fair.
Reliability	An investigation is reliable if it gives the same results when it is repeated.
Validity	An investigation is valid if it tests what it is supposed to test (a fair test).
Puberty	The stage when the body matures and becomes able to reproduce.
Menstruation	The monthly breakdown and loss of the uterus lining through the vagina.
Ovulation	The release of a mature egg from an ovary.
Fertilisation	The joining of a sperm cell and an egg cell to form a zygote.
Zygote	A fertilised egg cell.
Implantation	When the embryo attaches to the lining of the uterus.
Placenta	The organ through which the mother and the developing baby exchange food, oxygen and waste.
System	A set of parts that work together and affect each other.
Atmosphere	The layer of gases surrounding the Earth.
Hydrosphere	All the water on Earth, in liquid, solid and gas form.
Lithosphere	The rigid outer layer of the Earth: the crust and the upper part of the mantle.
Biosphere	All living organisms on Earth and the places where they live.
Rock cycle	The continuous process in which rocks are formed, broken down and changed from one type into another.
Igneous rock	Rock formed when magma or lava cools and hardens.
Sedimentary rock	Rock formed when layers of sediment are compacted and cemented together.
Metamorphic rock	Rock formed when existing rock is changed by heat and pressure.

Mineral	A naturally occurring, non-living solid with a definite chemical composition.
Ore	Rock that contains enough of a valuable mineral to be mined profitably.
Refining	The process of purifying a mineral or metal by removing unwanted substances.
Smelting	Heating ore to a high temperature to separate the metal from it.
Troposphere	The lowest layer of the atmosphere (0–12 km), where weather happens and temperature decreases with height.
Stratosphere	The layer of the atmosphere from 12–50 km that contains the ozone layer; temperature increases with height.
Mesosphere	The layer from 50–80 km where meteors burn up; the coldest layer.
Thermosphere	The layer from 80 km upwards where temperature increases greatly; aurora and satellites.
Ozone layer	A layer of ozone gas in the stratosphere that absorbs harmful UV radiation from the Sun.
Greenhouse effect	The warming of the Earth because greenhouse gases in the atmosphere trap heat given off by the Earth's surface.
Greenhouse gases	Gases that trap heat in the atmosphere, e.g. carbon dioxide, methane and water vapour.
Global warming	The increase in the Earth's average temperature caused by the enhanced greenhouse effect.
Star	A huge ball of hot, glowing gas that produces its own light and heat by nuclear fusion.
Nebula	A huge cloud of gas and dust in space where stars are born.
Nuclear fusion	The joining of small atoms (hydrogen) to form bigger atoms (helium), releasing a lot of energy.
Main-sequence star	A stable star that is fusing hydrogen into helium in its core.
Red giant	A large, cool, red star formed when an average star runs out of hydrogen in its core.
White dwarf	The small, hot, dense core left after an average star dies.
Supernova	The huge explosion of a massive star at the end of its life.
Black hole	What is left after a very massive star collapses; its gravity is so strong that not even light can escape.